

Khalid Mohamed Ekbal

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Aspiring Mechatronics & Automation Engineer with hands-on experience in embedded systems, control, Ev powertrain, and 3D printing Prototype .

Education

Bachelor of Engineering in Mechatronics & Automation, Ain Shams University

Expected 2027

Internships

Research & Development (R&D) Intern | El Sewedy Electrometer

07/2026 - Present

- **Embedded Firmware & Hardware Design:** Analyzed smart meter architectures, focusing on RTOS implementation, low-power MCU optimization, and industrial communication protocols (RF, GPRS, PLC), alongside ultrasonic flowmeter integration and PCB design principles (DFM/DFT, EMC compliance).
- **Systems Testing & Product Lifecycle:** Mapped end-to-end product development from 3D CAD mechanical modeling to full-stack software integration (ASP.NET Core, Angular), while applying ISTQB-aligned testing frameworks for REST APIs, performance load testing, and system validation.

GB Academy | Automotive Mechanical Advanced Trainee

08/2024 - 09/2024

- Completed training on engine operations, transmission, and braking systems.
- Acquired practical knowledge of key automotive components through practical sessions.

Experience

Freelance 3D Printing Business | Founder & 3D Printing Specialist

09/2025 – present

- **Prototyping & Client Consultation:** Designed and manufactured custom prototypes tailored to precise customer specifications and functional requirements.
- **Design for Additive Manufacturing:** Modeled and modified parts using SolidWorks, strategically designing around 3D printing limitations to ensure structural integrity and manufacturability.
- **Production & Material Selection:** Successfully managed a production volume of over 1,000 printing hours, expertly matching client mechanical and environmental needs with the most suitable printing materials.
- **Process Optimization:** Calibrated advanced slicer settings to optimize print quality, minimize failure rates, and reduce material waste.

ASU EV Racing Team| Ev powertrain member, Ain Shams university

11/2024 – 05/2026

- **Battery Systems & Safety:** Studied EV battery pack configurations, cell interconnections, and fuse protection strategies to ensure reliability and safety.
- **Battery Management Systems (BMS):** Gained knowledge of suitable BMS architectures for monitoring, balancing, and protecting battery cells.
- **Modeling & Testing:** Built MATLAB models for battery sizing and conducted tests to determine capacity, efficiency, and dynamic behavior of the battery.

Projects

AMR Medical Robot (Autonomous Mobile Robot)

- **System Integration & Control:** Designing a semi-autonomous medical robot with complete mechanical, electrical, and software integration. Implementing ROS 2 on a Raspberry Pi for high-level navigation, decision-making, and state machine execution.
- **Embedded Architecture:** Developing RTOS-based low-level control on an ESP32, utilizing PID motor control and sensor fusion (IMU and encoders) to execute dynamic obstacle avoidance and autonomous navigation.
- **Hardware & UI:** Engineered the complete power system architecture and safety circuits (E-stop, protection systems). Developed a vision module and Wi-Fi-based GUI for real-time telemetry visualization and supervisory control.

Automated Production Line (PLC-Based Control)

- **Automation & Logic:** Designed and implemented a comprehensive industrial production line encompassing feeding, machining, assembly, and sorting processes.
- **PLC & HMI Integration:** Programmed a Siemens S7-1500 PLC with structured automation logic for coordinated system operation. Developed an HMI interface for real-time monitoring, control, and alarm management.
- **Simulation & Safety:** Implemented fault detection, system safety logic, and adaptive conveyor control based on system load, validating performance through SIL/HIL simulation in Factory I/O.

7-DOF Robotic Arm (ROS 2 & Micro-ROS)

- **Kinematics & Motion Planning:** Designed and developed a 7-degree-of-freedom robotic manipulator, implementing a ROS 2-based motion planning and control architecture for coordinated, multi-joint movement.
- **Distributed Control System:** Integrated micro-ROS communication between the high-level ROS 2 system and embedded controllers. Developed RTOS-based low-level actuator control to ensure real-time performance and synchronized joint stability.

Smart Flowmeter System: Designed a high-accuracy (92%+) cup-anemometer-based flowmeter from the ground up, including parametric modeling in Excel, complete 3D CAD assembly, and calibration methodology. Independently developed the IoT system logic and sensor integration, while contributing to CAD design, sensor placement, and workshop assembly to ensure accurate and manufacturable flow measurement.

Courses

Embedded Systems Course -Robotics Corner (2025)

Ev Powertrain Academy –ASU racing team(2024)

Machine Learning Course - StepsTowardsProgress (2024)

Skills

- **Programming& Firmware:** C, C++,Python, Embedded C, Modern C++, Arduino, Machine Vision.
- **Simulation & CAD:** MATLAB, SIMULINK, Ansys, Autodesk Inventor, Fusion 360,SolidWorks.
- **Productivity & Design:** MS Excel, MS Office
- **Languages: Arabic:** Native | **English:** Intermediate.